

RFID Side Channel Analysis for IoT Devices

Katherina Keogh
MEng. Electronic and Computer
Engineering,
Dublin City University,
Dublin, Ireland
katherina.keogh24@mail.dcu.ie

Abstract—The rapid increase of Radio Frequency Identification (RFID) technology within the Internet of Things (IoT) ecosystem has revolutionised applications ranging from inventory management to healthcare. However, alongside its benefits, RFID systems face critical security vulnerabilities, particularly in the context of side-channel analysis (SCA). This paper investigates the susceptibility of RFID systems to side-channel attacks, which exploit unintended physical emissions such as power consumption and electromagnetic radiation to extract sensitive information. Supported by prior research and analysis, this work highlights key security challenges, focusing on vulnerabilities in RFID protocols and their implications for privacy and data integrity. The potential of enhancing side-channel analysis techniques using tools such as the Proxmark 3 proposing firmware modifications to improve its utility in evaluating RFID security is also discussed.

Index Terms— Internet of Things, Proxmark, Radio Frequency Identification, Side Channelling

I. INTRODUCTION

RADIO Frequency Identification (RFID) technology has become an important part of the Internet of Things (IoT) ecosystem, enabling communication and identification in applications ranging from supply chain management to healthcare and smart cities [1]. As RFID systems continue to grow, their security and reliability have lacked significant attention, particularly in protecting sensitive data and ensuring operational integrity [2]. Despite advancements in encryption and authentication protocols, RFID systems remain susceptible to various vulnerabilities, including side-channel attacks, which exploit unintended information leaks to compromise security [3]. Side-channel analysis (SCA) is a growing area of concern in RFID security, using signals such as timing information, power consumption, and electromagnetic emissions to extract confidential data [3]. These non-invasive attacks bypass traditional cryptographic defences, making them hard for RFID systems to detect. Given the increasing deployment of RFID in security-critical applications, understanding and reducing the risks posed by SCA is essential to enhancing IoT device security.

This review explores the RFID technology and side-channel analysis, focusing on identifying vulnerabilities and evaluating the potential of tools like the Proxmark 3 to display their weaknesses.

II. RFID IN IOT

An RFID (Radio Frequency Identification) system is a wireless communication technology for automatically identifying and tracking objects, animals, or people. It consists of three main components: a tag, a reader, and a backend system. The tag, which can be either passive, active, or semi-active, contains a microchip for storing data and an antenna for transmitting that data to a reader. The reader emits radio frequency signals to power and communicate with the tag in the case of passive tags or to receive data from active tags that have their own power source. The backend system processes the data received by the reader and integrates it into applications such as inventory management, access control, or supply chain logistics. RFID systems operate at various frequency ranges, such as low frequency (LF), high frequency (HF), and ultra-high frequency (UHF), depending on the use case. Their ability to enable contactless communication over short distances makes them popular in numerous industries.[4]

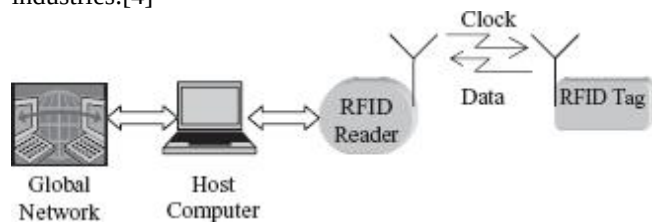


Fig. 1. Block diagram of a typical RFID system. [4]

In practical applications, RFID has transformed industries by enabling real-time asset tracking, automation, and data collection. Retail giants like Wal-Mart leverage RFID for efficient inventory management [4], while the healthcare industry relies on it for equipment tracking and patient identification. Transportation systems integrate RFID in toll collection and contactless fare cards, and smart cities use the technology to monitor infrastructure and deploy IoT solutions. These implementations highlight RFID's ability to enhance efficiency and precision in complex operations.

Despite these advancements, several underlying aspects of RFID systems can introduce challenges. The reliance on radio frequency signals, while enabling continuous communication,

also creates opportunities for interception or disruption of data transmissions. In applications requiring low-cost tags, the computational limitations of passive tags often restrict their ability to implement robust cryptographic protocols [5]. These characteristics make RFID systems attractive targets for malicious actors seeking to exploit authentication or data protection vulnerabilities. As RFID systems become increasingly integrated with IoT networks, the consequences of security weaknesses are amplified, demanding significant research efforts to enhance the resilience of this critical technology.

III. RFID VULNERABILITIES

RFID technology, while revolutionising industries with its data transmission capabilities, faces significant security challenges. Side-channel attacks (SCAs) represent a critical and often underappreciated threat among the numerous attack vectors. SCAs exploit physical characteristics and unintended emissions of RFID devices rather than directly targeting the underlying communication protocols.

RFID systems are vulnerable to side-channel attacks due to their constrained computational resources and reliance on lightweight cryptographic protocols. Unlike traditional attacks such as eavesdropping or cloning, SCAs analyse physical emissions, such as power consumption, electromagnetic radiation, and timing information, generated during the operation of RFID tags or readers [3]. For example, during cryptographic operations like encryption or authentication, variations in power usage or electromagnetic emissions can leak sensitive information, including cryptographic keys. This vulnerability arises because many RFID devices are designed to prioritise cost and efficiency over physical security.

A. Cloning RFID Cards

The study carried out by Jain, Chaudhary, et al. underscores the inherent vulnerabilities in RFID technology, particularly in its current state of implementation. The ability to easily access, alter, or clone data stored on RFID tags reveals significant weaknesses that can compromise privacy and security. The research highlights the susceptibility to denial-of-service attacks, where card data can be maliciously overwritten, further highlighting the fragility of these systems. The experiments performed on a University Identification card and blank cards demonstrated how effortlessly such attacks can be executed, reinforcing the urgent need for advancements in RFID security. [2]

B. Hamming Distance Leakage Model

The study carried out by Xu, Zhu, et al. further emphasises the critical vulnerabilities inherent in RFID and smart card systems when subjected to side-channel attacks. By using Hamming distance leakage and targeting specific rounds of DES encryption, the research highlights how sophisticated combinations of attack methods can bypass existing security measures [3]. This highlights the inadequacy of traditional countermeasures such as "head and tail" protection, which fail to prevent attackers from exploiting subtle leaks in the cryptographic process. These results not only demonstrate the possibility of side-channel attacks but also point to the need

for enhanced defensive strategies to reduce these risks and ensure the integrity of sensitive data in RFID systems.

C. Reverse Engineering

The study carried out by Fraga-Lamas and Fernández-Caramés further emphasises the importance of securing RFID systems. In this study, the authors reverse-engineered the communication protocol of an RFID public transportation card. They demonstrated how an attacker could manipulate the system to gain unauthorised access and clone the RFID card. This research highlights the vulnerability of RFID in real-world scenarios, particularly in public-facing applications such as transportation, where attackers can exploit communication protocol weaknesses to compromise the system.[6]

D. Machine Learning-Based Side-Channel Attacks

Recent advancements in side-channel analysis have introduced machine learning (ML) as a state-of-the-art approach for attacking cryptographic systems. These techniques use ML algorithms to model and exploit physical leakage, such as power consumption or electromagnetic radiation, during sensitive computations. Unlike traditional methods, machine learning-based side-channel attacks are especially powerful in scenarios where only a limited number of profiling traces are available. By combining machine learning methods with side-channel analysis, researchers have identified new ways to bypass the constraints of classical approaches like template attacks (TAs) [7].

To validate these findings, researchers conducted experiments using a novel metric called the Data Confusion Factor (DCF), which measures the success of machine learning models in profiling and attack phases. These experiments showed that ML-based side-channel attacks achieved higher success rates than TAs, especially in cases involving high-dimensional traces with many irrelevant samples. The use of dimensionality reduction techniques, such as principal component analysis (PCA), has further enhanced the efficiency of ML-based attacks by focusing only on the most informative features of the leakage traces [7].

IV. SIDE-CHANNEL ANALYSIS ON RFID RESEARCH

Side-channel analysis (SCA) is a powerful tool for evaluating vulnerabilities in RFID systems. SCA exploits physical information leakage, such as electromagnetic (EM) emissions, power consumption, or timing discrepancies, to infer sensitive data without directly attacking cryptographic algorithms [3]. Side-channel attacks reveal critical weaknesses in RFID systems by analysing physical manifestations of computations. Differential power analysis (DPA), for instance, can extract cryptographic keys by comparing power consumption across multiple operations, while timing attacks deduce sensitive data based on processing delays. Electromagnetic attacks are particularly relevant to RFID due to the wireless nature of its communication, allowing EM emissions to be intercepted and analysed. [8]

The Proxmark is a versatile, open-source tool designed for the analysis, manipulation, and emulation of RFID systems [9]. The device is widely used in security research, enabling tasks such as reading and writing RFID tags, capturing communication data, and performing advanced attacks like

cloning, eavesdropping, and replay attacks. With its customisable firmware and powerful hardware capabilities, the Proxmark is a critical platform for investigating vulnerabilities in RFID systems and testing potential countermeasures.

To maximise Proxmark's efficacy for SCA, several firmware modifications can be made.

A. High-Resolution Signal Capture

The current functionality of the Proxmark device captures communication signals from RFID systems, but it may lack the granularity required for detailed side-channel analysis. As stated, side-channel attacks often rely on detecting subtle variations in physical signals such as EM emissions or power consumption, meaning these variations can reveal sensitive data, including cryptographic keys or sensitive information being processed by RFID devices.

The firmware could be enhanced to address this limitation and support high-resolution sampling and data storage. Modifications can allow for a more precise analysis of the physical characteristics of RFID communication, improving the ability to detect leaks in physical emissions. For example, the `client/src/cmdhf.c` [10] file, which handles high-frequency commands, could be modified to introduce a high-resolution capture mode. By enhancing this section of the firmware, the Proxmark would be better equipped to collect and store detailed EM data, enabling fine-grained analysis of physical leakage. This capability would facilitate more accurate timing, power, and electromagnetic analysis, which are crucial for identifying vulnerabilities in RFID systems.

B. Enhanced Timing Analysis Tools

Timing analysis is a common technique in side-channel analysis, relying on precise measurements of processing delays and computation timings. Many side-channel attacks, such as timing attacks, exploit these processing delays to infer sensitive information, such as cryptographic keys or data patterns. These delays are often generated during cryptographic operations in RFID tags or readers.

Upgrading the Proxmark firmware to log timestamps with higher accuracy could significantly improve the reliability of timing-based attack experiments. For example, the file `client/src/cmdhf14a.c` handles high-frequency ISO14443A commands, which are often used in RFID communication protocols [10]. By modifying this file to increase timestamp resolution, researchers could accurately capture the precise timing of events within the system. This would allow researchers to observe even small delays, which could reveal vulnerabilities in cryptographic operations or authentication protocols.

With this enhancement, researchers could detect variations in processing times that would otherwise go unnoticed with lower-resolution timestamp logging. These subtle variations can be exploited in timing-based side-channel attacks to obtain cryptographic keys or to reveal patterns in data transmission. For example, side-channel analysis could reveal information such as whether a tag is performing encryption, the timing of key generation, or whether certain cryptographic steps are skipped during processing.

Through these proposed firmware modifications, the Proxmark would provide a secure platform for conducting more effective and detailed timing-based side-channel analyses in RFID systems. These tools would not only validate existing research but also explore vulnerabilities that could be exploited in practical scenarios.

V. PRELIMINARY WORK

To establish a functional environment for conducting the research, preliminary setup steps were carried out on a macOS machine. The essential development tools, including Apple's XCode Command Line Tools, which provide functions like `git` and `make` for building software, had already been previously installed and set up. The Proxmark 3 Iceman firmware repository was then cloned from GitHub using `git`, enabling access to the latest firmware updates and modifications. Homebrew, a widely-used package manager for macOS, was used to install dependencies such as `gcc`, `libusb`, and `cmake`, ensuring the Proxmark client and firmware could be compiled and executed seamlessly.

The next step will involve downloading the Proxmark client software and following detailed instructions to flash the firmware onto the Proxmark 3 hardware. This will require connecting the device to the Mac via USB, ensuring compatibility with the device's serial drivers. Specific configuration files within the firmware repository, such as `cmdhf.c` and `cmdhf14a.c`, were examined to identify potential areas for customisation. Testing the setup will involve running basic Proxmark commands to verify communication between the software and hardware.

VI. CONCLUSION

The security of RFID systems remains a critical concern as these technologies become increasingly essential in the Internet of Things (IoT) ecosystem. This paper has examined the vulnerabilities of RFID systems, with a focus on their susceptibility to side-channel attacks (SCAs). By exploiting physical emissions such as electromagnetic emissions, timing variations, and power consumption, SCAs bypass traditional cryptographic protections, exposing sensitive data to malicious actors. These attacks are particularly concerning given the constrained resources of RFID tags, which often prevent the implementation of secure security measures.

Prior research has demonstrated the possibility and effectiveness of SCAs in exposing weaknesses in RFID systems. Studies such as those by Jain et al. and Xu et al. have highlighted the ability of attackers to extract cryptographic keys and manipulate data through methods like Hamming distance leakage analysis and differential power analysis. These findings underline the inadequacy of existing countermeasures and the urgent need for advanced security strategies to protect RFID systems.

The use of tools like the Proxmark 3 offers significant potential in identifying and reducing vulnerabilities. The ability to capture high-resolution signals and conduct detailed timing analysis makes the Proxmark an invaluable resource for evaluating the security of RFID systems. Proposed firmware enhancements, such as increasing signal capture

resolution and improving timing precision, would further strengthen its side-channel research and testing capabilities.

As RFID continues to integrate with critical applications in supply chain management, healthcare, transportation, and smart cities, ensuring the security and resilience of this technology is increasingly important. This research highlights the need for continued research in defensive measures, such as side-channel-resistant cryptographic protocols and improved hardware designs, to further reduce risks. Addressing these challenges is essential to maintaining and protecting sensitive data.

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